Zachary Petersen

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EDUCATION

University of Wisconsin – Madison GPA: 3.44/4.0

Nuclear Engineering B.S. Spring 2027, Astronomy–Physics Major Spring 2027, Physics Major, Spring 2027

EXPERIENCE

2026 Nuclear Security & Nonproliferation Summer School – June 9-12, 2026 at LLNL hosted by UC Berkeley

Lawrence Livermore National Laboratory – Defense Science and Technology Internship (pending background investigation) with mentors Vladimir Mozin, Brandon Clary, and Spike Procassini, Summer 2026

- Support strategic deterrence and survivability efforts at the laboratory

Department of Nuclear Engineering – University of Wisconsin – Madison – Research for Sebastien Philippe, Fall 2025 – Present

- Modeling the damaging effects of High-Altitude Nuclear Explosions (H.A.N.Es) on satellites and the Van Allen Belt for a paper in the United Nations Panel on Effects of Nuclear War

Lawrence Livermore National Laboratory – Defense Science and Technology Internship with mentors Vladimir Mozin, Brandon Clary, and Spike Procassini, Summer 2025

- Used MCNP to model the Dosimetry System dose calculations of the war time bombings of Hiroshima and Nagasaki to validate the LLNL Survivability Group findings. This work supported strategic deterrence and survivability efforts at the laboratory.

Oak Ridge National Laboratory – Summer Undergraduate Laboratory Internship under Walid Metwally, Summer 2024

- Used MCNP to perform shielding and criticality calculations for HALEU Transportation Canisters. Investigated various source definitions and perturbations to the canister for design optimization.

Office of Radiation Safety - University of Wisconsin – Madison – Radiation Safety Technician, August 2022 – Present

- Survey campus labs that use radioactive material and run vials through a Liquid Scintillation Counting Machine (LSC) to determine if the labs are safe. Provide General Help to Health Physicists and Office of Radiation Safety Staff

Department of Physics – University of Wisconsin – Madison – Research for Mark Friesen, Spring 2023 – Spring 2024

- Used Python to model electron valley splitting in quantum qubit devices with the objective to optimize the quantum well to enhance valley splitting

SKILLS

- Computing: Python, VisIt, R-Studio, MATLAB
  - 2+ years of experience with High-Performance Computing Systems and parallel programming
- Radiation Transport Codes: MCNP6, SCALE, (limited Mercury)
- Certified Radiation Worker – Certificate of Training and Completion (09/23- Present)

AWARDS

- Physics Department Promising Future in Physics Student Award: Hagengruber Undergraduate Support Fund
- College of Engineering Scholarships:
  - The Dorothy Marshall Frawley to honor Charles J Marshall Frawley & Mensel, Robert J Scholarship
- Department of Nuclear Engineering & Engineering Physics Scholarship
- Nuclear Regulatory Commission Scholarship - 2024 and 2025